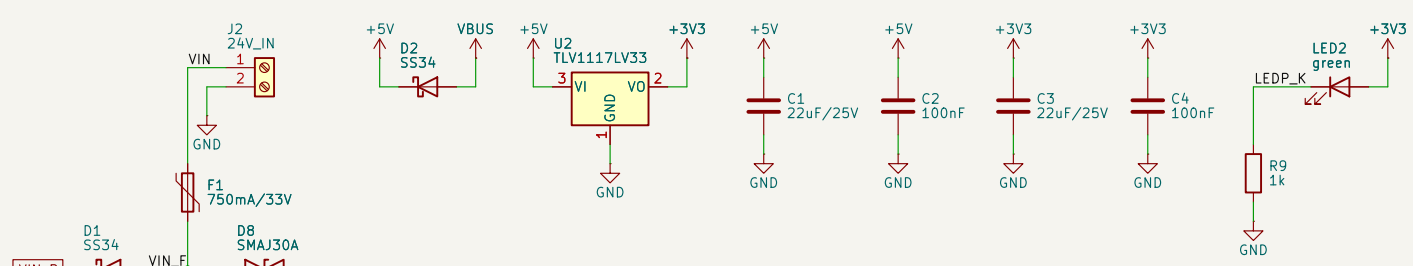
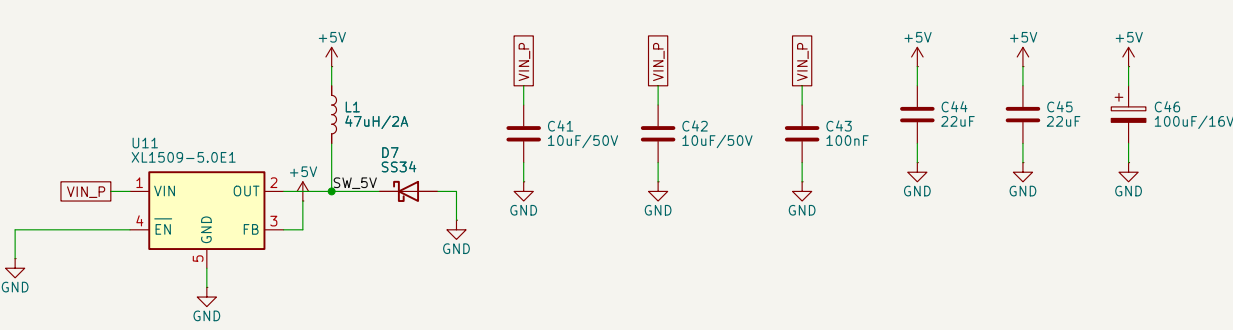


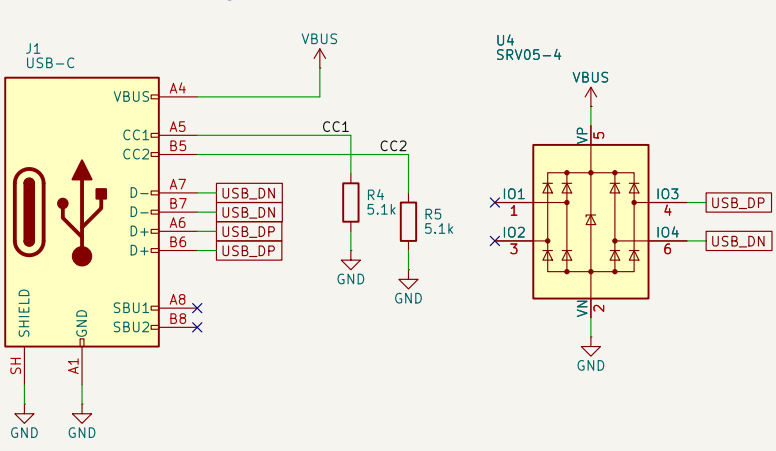
POWER IN: 24 VDC terminal or USB
F1+DB clamp-and-blow at the entry: D1 blocks reverse polarity.
U11 (right) makes +5V; D2 ORs in USB VBUS; U2 drops it to +3V3



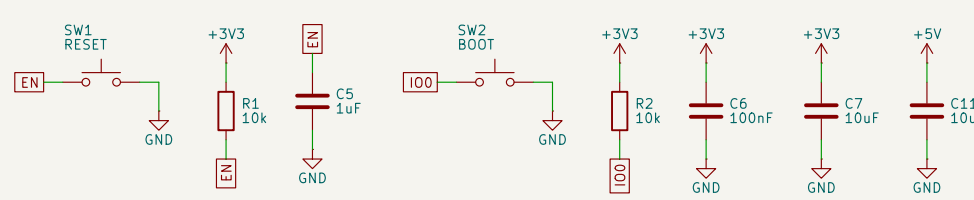
24 V -> 5 V BUCK XL1509-5.0, fixed output
C41-C43 in, D7 catch, L1 47uH, C44/C45/C46 out. FB ties to the output because the part is the fixed 5.0 V version; EN is active LOW and grounded. C46 is an ELECTROLYTIC on purpose: voltage-mode with fixed compensation needs its ESR zero, and an all-ceramic output has none. Keep C41/C42 - U11 - D7 tight: that loop, not L1, is what radiates.



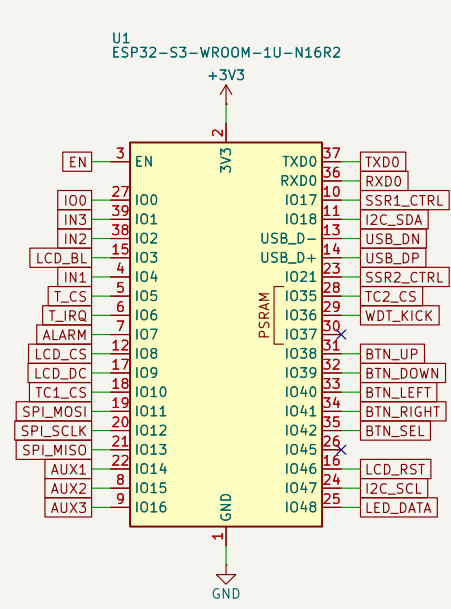
USB-C native USB flashing + ESD



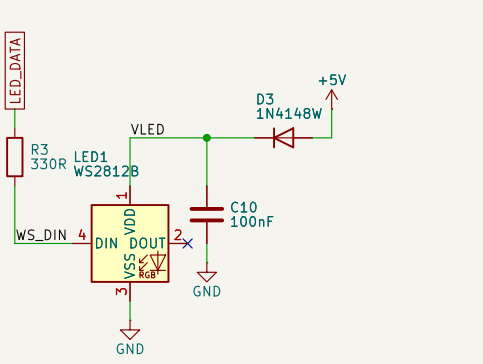
RESET / BOOT / DECOUPLING



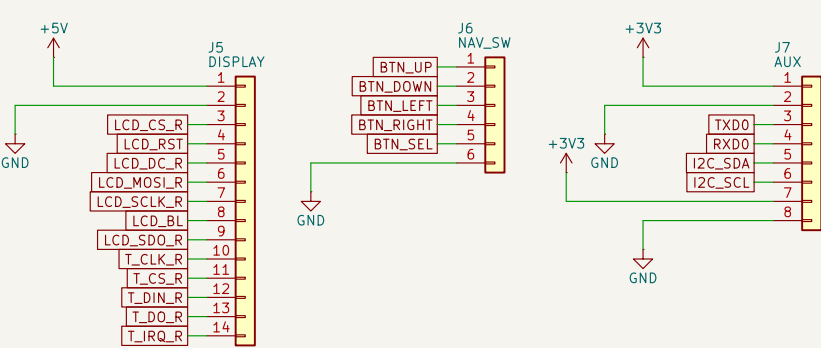
ESP32-S3-WROOM-1U-N16R2
GPIOs = firmware Kconfig defaults



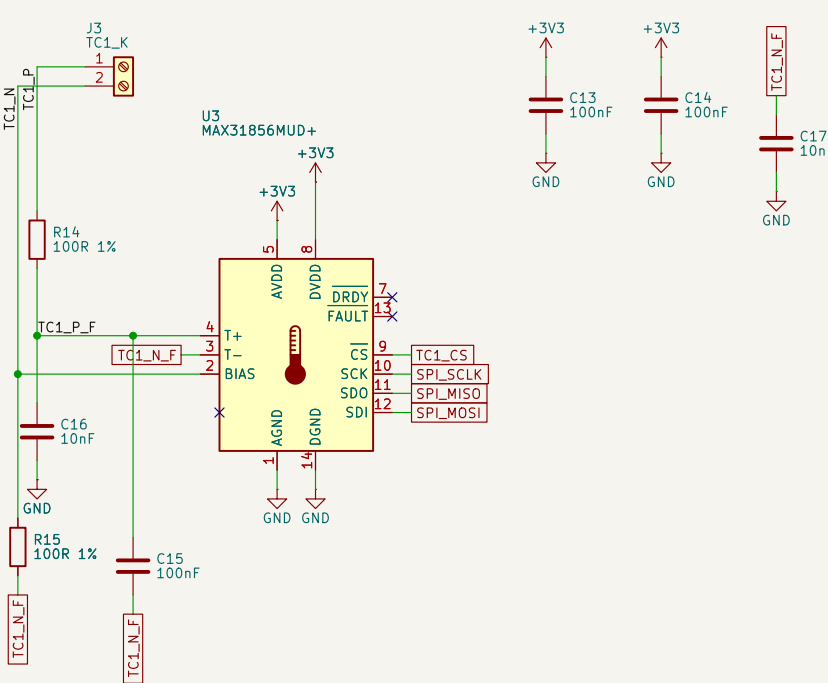
WS2812B STATUS LED
VDD dropped -4.6V for 3.3V data margin



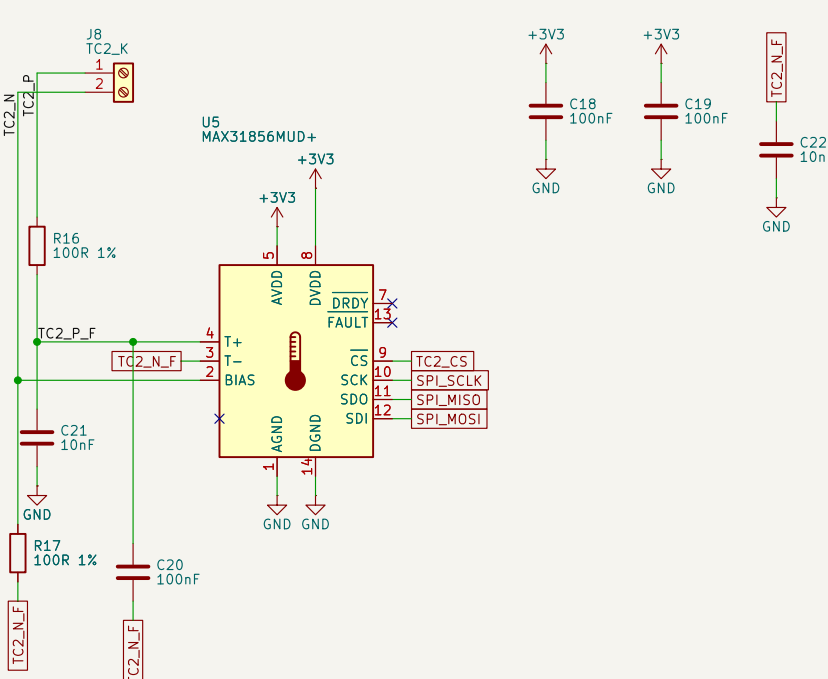
HEADERS
J5 display+touch (14-pin), J6 nav, J7 aux+I2C



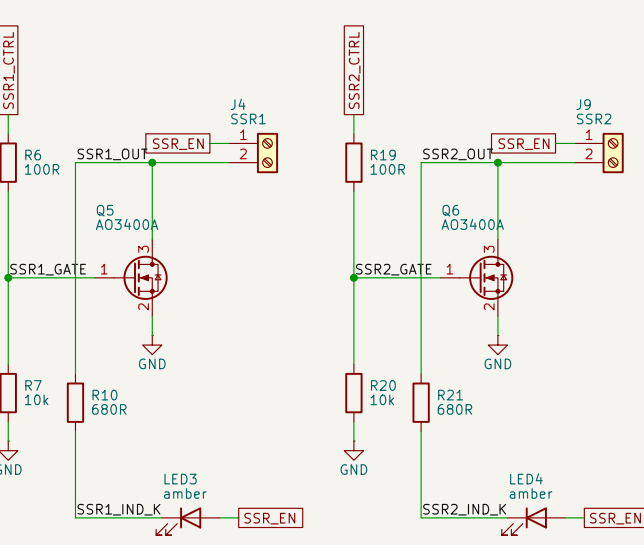
THERMOCOUPLE 1 (control) MAX31856
T- floats to J3, biased via BIAS



THERMOCOUPLE 2 (load) MAX31856
T- floats to J8, biased via BIAS



SSR DRIVE x2
low-side A03400A; indicator LED across each terminal



HARDWARE WATCHDOG
U10 retriggerable one-shot: every rising edge on WDT_KICK restarts a ~2.2 s window, so Q stays high only while firmware keeps kicking. Q holds Q3 on; Q3 pulls SSR_PG down, turning on high-side Q4, which supplies SSR_EN - the +5V rail feeding BOTH SSR terminals and both indicators.
R46/C38 set the window. R48 holds B low when the MCU is high-Z. R47 is the fail-safe pull-up. S/J2 = bring-up defeat, REMOVE for service.

